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The modernist spolia

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ABSTRACT: This paper investigates and broaden the concept of *spolia* in connection to buildings from the industrial period to discuss, how a modern concept of *spolia* encourage a circular economy by preserving materials, constructions, and questions what kind of architectural strategies can be developed from this?

To analyse and discuss these matters, this paper takes point of departure in a case study of the renowned project Resource Rows (*Resourcerækkerne*) by the architectural firm Lendager. The project is analysed through the concept of *spolia* as described by Maria Fabricius Hansen (1964) and the tectonic theory on *stofwechslen* as established by Gotfried Semper (1803-1879).

1 INTRODUCTION

1.1 *The modernistic building stock and the new green agenda*

The International Plant Protection Convention (IPPC) suggested in 2023, that one of the many strategies of mitigating the climate crises is to reuse and transform existing buildings in a larger scale, than practise is today (IPPC 2023). In a Danish context, the philanthropic organisation, Realdania, has suggested, that the buildings from the industrial period from 1930-1970 play a significant role when it comes to reusing the building stock (Petersen et al. 2020). In this period the use of industrialized building practise gave rise to rational industrial districts, large mass housing projects around the major cities and the development of single-family houses. As this period represents a large part of the overall building stock in Denmark, the materials also come with an imbedded value representing a high amount of embodied energy (Petersen et al. 2020). This makes the refurbishment of the buildings and – if not used in its whole - reuse of components and materials from them, an important part of developing a more sustainable building practice.

2 METHOD AND THEORY

2.1 *Method*

The study is based on in-depth analysis of one case. Information of the case is collected in three levels. Firstly, a site visit was executed, experiencing the building physically and visually through first hand experiences as well as photo documentation. Secondly, written and drawn sources were studied. On this basis the the third level relates to future perspectives of the case, where central architects from Lendager with special knowledge to the project was interviewed.

The theory used to structure the tectonic analysis of the case is based on the core-notions of Gotfried Sempers theory on stofwechslen. The analysis forms the basis of a discussion of the case as a modernist spolia.

Although the case focus on a unique way of recycling bricks from modernised buildings, it is still an example of general trends of recycling within a Danish context. This makes it possible to use the case to both discuss a specific new solution and to raise a more general discussion on how we should develop the field in the future.

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2.2 *The concept of spolia*

Throughout history people have used and reused buildings, materials and building components as the most elementary way of acquiring sturdy, usable or scarce resources and keeping the economic (and time-consuming) cost low. Multiple historic examples show that this practice has been used across all social classes. Anecdotal and documented reuse traditions are registered in the vernacular architecture, where timber construction was commonly replaced both within the building and in other buildings, which is showed by scrambled numbering marks and out-of-place holes for jointing (Engquist 1989; Kirk 1980). But also, in manors and castles sturdy materials were harvested from other buildings and reused. E.g. In the 14th century manor of Boller (Horsens, Denmark), building materials from a medieval church were reused, when the political authority changed from the church to the nobilities during the 17th century - probably both for convivence, but also as a demonstration of power (Andersen 1996).

On the notion of reusing materials as a power demonstration the Danish art historian, Professor Maria Fabricius Hansen defines spolia as an integrated architectural practise in the medieval Rome (Fabricius-Hansen 2010). It might have originated in an imperial desire to implement cultures or symbols from other places or ages, and it seems like, that in the through time reuse became highly appreciated and elements were even composed to be as motely as possible as it became fashionable symbols of power, spiritual cohesiveness, and luxury (Fabricius-Hansen, 2010). Fabricius- Hansen suggests, that spolia both had an ideological and a physical (practical) connotation in antient Rome:

"Salvaging columns from older contexts and making sure that their ancient origin remained visible through the heterogeneity of the total result was a way of referring to the past and letting it work as a literal and ideological foundation in the present." (Fabricius-Hansen 2003 p. 151)

The tradition of reusing building materials and –components have been a precondition throughout history. But the urge to 'exhibit' the reused materials has varied through time and only during the industrialisation of the last century, possibly in the unusually stable period in western society after second world war, reuse has almost been irradicated.

2.3 *Gottfried Semper and the concept of stoffwechsel*

Architect and art historian, Gottfried Semper (1803-79) contributed immensely to the foundation of tectonic theory as we know it today. An important part of Sempers' research was on the development of architecture through history in order to show the evolution of archetypical elements, use of materials, building techniques, types of dwellings and spatial motives from various cultures. From this Semper conceived and applied the term *stoffwechsel* through which he explains the evolution of architectures constituent parts – and of all manmade artefacts for that matter. S*toffwechsel* can be used to describe the way one structural form - a motif - which originates from one material and a method of processing pertaining to that material, can be transferred or applied to another type of material and/or method of processing, thereby releasing the form from its original cultural connotations. Of this, Semper writes in *Style:*

"If the motive undergoes a new change of material [Stoffwechsel] as a result of this secondary or even multiple transformation, the resulting new form will be a composite, one that expresses the primeval type and all the stages preceding the latest form." (Semper & Mallgrave 2004 p. 250)

Following Sempers theory, *stoffwechsel* takes place through substitution of materials and technological developments of bringing forth archetypical artefacts. (Ejstrup 2019) The complexity of contemporary construction with regards to its use of composite materials blurs the precision of Sempers' evolutionary tracking of forms and the materials they are made of. However, the archetypical constituent parts of architecture remain, and from this it is interesting to introduce the concept of recycling, where materials and components are used in different ways than originally intended. Recycling of composite materials through contemporary construction methods could be seen as a further evolution within the understanding of *stoffwechslen.* As such Semper's research on this topic shows *stoffwechslen* as a theoretical framework for a continued reflective evolution of architecture. (Moravánszky 2017).

3 CASESTUDY: RESOURCE ROWS

3.1 *Introduction to the case*

The 'Resource Rows' development was completed in 2019 and consists of 92 rental homes (29 terraced houses and 63 apartments). The building is located in Ørestad Syd, which is part of a new urban area close to the centre of Copenhagen. The project is designed by the architectural office Lendager for the developer NREP. As the name suggests, the design of the development has a strong focus on reducing resource consumption through the use of recycled materials in the construction. This innovative approach to construction has earned the building much publicity in Danish and international media and in 2022 it was nominated for the Mies van der Rohe Award (Lendager & Pedersen 2020). This case study will focus solely on the building's brick facades facing the street and not on the other examples of the use of recycled materials in the construction.

3.2 *The concept of a façade of recycled brick fragments*

Recycling of bricks has been an integrated part of the building culture for centuries. In a Danish context we have numerous examples of bricks being harvested from old disused or burnt buildings and used in new structures. What makes the façade of the Resource Rows special is the fact that the recycled bricks are from resent buildings completed after 1960. Before this period, brick facades were built with lime mortar, which made it possible to disassemble the facade and clean the individual bricks. From 1960 onwards, cement-based mortar began to be used, which is stronger than the brick itself. This makes it very difficult to take down a brick façade without destroying the bricks and calls for new ways of working with the material. The brick facade in this case is based on a new strategi of reclaiming bricks as larger fragments developed by the architects at Lendager.



Photo 1. Images of the process of cutting the brick fragments and the final brick wall. Photo: Lendager.

To avoid destroying the bricks during dismantling, the facade is manually cut into 1 x 1 meter squares with a diamond blade. These brick fragments are lifted off the facade and fixed with wooden frames so that they can be transported to a manufacturer of precast concrete elements. At the factory the brick fragments are placed in the casting moulds for the concrete elements and a concrete back wall is cast over the fragments. In this way the brick fragments become the outer wall of a sandwich wall panel. The fragments are fixed to the concrete wall and therefore have no load-bearing function in the façade (Pasternak 2019). This creates great freedom in the positioning of the elements in relation to each other, allowing for a rich variation in the expression of the facade. This facade system was developed specifically for this project and is so far the only example of this type of facade in a Danish context.



Photo 2. The south façade of the Resource Rows. Photo: Lendager.

3.3 *The future development of the concept of the recycled brick facade*

One of the disadvantages of the current building system is the need for a heavy concrete back wall. In terms of a more sustainable use of resources, both the concrete wall and the energy consumption for dismantling the brick fragments pose a challenge. To meet these challenges, the office of Lendager has developed a new version of the wall system called ReWall where the concrete back wall is replaced with a lightweight construction of recycled wood. This system has been developed as part of Xan Browns industrial PhD-project (Brown 2024). In this system, the brick fragments are placed on steel brackets fixed to the wooden structure. The fragments are then secured to the structure with 4 bolts. This principle creates a very different expression in the façade. The brick fragments now appear as self-supporting blocks held by the steel brackets. In addition to saving resources by switching from a concrete back wall to a wood construction, the solution is also designed for disassembly. This ensures that the facade can be disassembled, and both the brick fragments and the wooden structure can be reused. The ReWall system is only a prototype so far and has yet to be tested in a construction project.



Photo 3. Mockup of the ReWall. Photo: Xan Brown.

4 CASE ANALYSES

4.1 *Brick – an essential part of the contemporary Danish building culture*

When looking at the area surrounding the Resource Rows in Ørestad Syd all the buildings are clad with bricks. This reflects a dominating trend in the contemporary Danish building culture. In the past decades new urban developments have been based on the same construction principles with a loadbearing system of concrete elements clad with a facing brick wall. These new urban areas in this way reflects the strong tradition of working with brick cladding in a Danish context.

Red tile bricks have been used in Denmark since the early Middle Ages, as the underground has been rich on red clay (Lorenzen, 1920). Historically it was the buildings of the clergy and nobilities that were constructed in bricks but during the 19th century, when the industrial production were introduced, bricks became more economical reachable for the broader public society and was implemented as a national symbol of good craftmanship and solid architecture during the movement of *Bedre Byggeskik* ('Better Building Practise') in the start 20th century (Slots- og Kulturstyrelsen, 2022). In the 1930ies international movement of modernism gave rise to the national interpretation, functionalism, which also tapped into the long tradition of building with bricks and made brickwork even more commonly during the next decades as large housing complexes, public institutions and small and medium factories and companies were built across the country in this style becoming a symbol of the Danish welfare state (Millech 1951).

In the architecture of today the ornamentation of the brick facade has become one the most important architectural elements of many buildings. The facing brick wall is not loadbearing, and it creates a freedom to work with different types of effects and ornamentation in the facade. The recycled brick façade of the Resource Rows can be seen as an expansion of the existing repertoire of ornamental brick expressions of the modern city. To understand the specific character of this new façade a tectonic perspective is needed.

4.2 *A tectonic perspective on the recycled brick façade*

When seen in a tectonic perspective the recycled brick façade of the Resource Rows represents a radical shift in the way bricks traditionally appear as part of a façade. We have already described how bricks has gone from being the primary load-bearing structure of a building to now only being the outer wall. The recycled brick façade takes this a step further as it is not a self-supporting outer wall, but rather a cladding of a concrete wall. Here, the façade's square brick fragments become a tectonic motif that is more akin to a tile cladding than to traditional masonry. The tile motif is emphasised by the way the architects deliberately used the tile fragments both in a vertical and horizontal position. This makes it clear that the brick is not load-bearing, as the wall courses are both cut and turned upside down. The bricks of the fragments are simple machine-made bricks, so it is the way the fragments are put together that creates variation in the facades.

In the light of Semper's ideas about stoffwechsel, the development of the brick motif can be seen as a transition from one type of construction principle to another. By cutting a brick wall into blocks that can be attached to a new facade, an ambiguity is created in the new facade expression. You can recognize the logic of traditional brickwork in the brick blocks, while a new type of facade structure is emerging. In the new façade, the masonry has lost its character as a load-bearing structure but has retained its qualities as a building envelope. The facade is characterized by the brick's weather-resistant appearance, which can withstand the humid climate in Denmark. The recycled brick façade of the Resource Rows both maintaining an image of traditional masonry and rethinking this image by using new construction principles. It challenges our notions of what a brick wall is and points towards a whole new way of working with masonry. The special logic associated with the use of brickwork in modernism becomes an essential element in the expression of the new facade. The fact that masonry from this period cannot be disassembled and therefore must be reused based on different principles

is the tectonic key to understanding this facade. Recycling becomes the main motif of the facade and thus its meaning is based on the historical references that the recycled material represents. You could say that the brick fragments become symbols of another time, a kind of a modernist spolia.

4.3 *The recycled brick façade as a modernist spolia*

The initial 'image' of the brick facade of Resource Rows is the concept of sustainability through recycling. But considering Marie Fabricius-Hansens concept spolia being material reuse as a referring to a past ideological foundation, in this case the modernistic tradition in Denmark, the facade also ideologically is connected to a broader historical social movement. By this, the reuse of the material becomes two-sided, as it both referrer to a literal statement of the present - sustainability trough recycling - but also to a deeper historical ideology tapping into the Danish welfare tradition of the mid-20th century which again refers to a thousand years old tradition of brickworks in Denmark – a tradition, which is broadly venerated by the citizens.

At the same time, the stofwechslen of the material from the traditional tectonic logic as a 'stacked wickerwork', to the use as a tile cladding demonstrates the technical problems of modern building techniques and reuse, as the bricks cannot be separated because of the concrete mortar. By this the tile motif becomes an ambiguous statement both referring to the ideology of the welfare state, criticizing the existing circumstances, where plenty of solid buildings are being demolished, and exhibiting the problematic building techniques of the late modernism.

On the other hand, the reuse of the bricks as a tile cladding has had an enormous consequence on the load-bearing structure in regards of environmental impact, which is hidden in the motif for the untrained observer.

4.4 *A critical perspective on the future of the brick façade*

Critiques of the consequence of the brick tile's used as a cladding on the lead-bearing structure of the Resource Rows have been publicly formulated even by the office of Lendager (Lendager & Pedersen 2020). Especially the design concepts are heavy regarding its high environmental impact. The second iteration of the construction, The ReWall project, tried to meet these challenges cladding the brick tiles on a lightweight timber load-bearing construction. This design sets the question if this further iteration on stofwechslen of the traditional, heavy brick construction, where it becomes a cladding on a lightweight construction, is even more tectonically unclear. Might reuse of inseparable bricks call for another kind of use than load-bearing structures or facade cladding? Furthermore, it questions if bricks can even be justified in the building practice of tomorrow unless it is the reuse of existing structures in its whole or materials, that can be easily disassembled thereby keeping their initial static and technical benefits?

The symbolic value of novel tectonic approaches and cultural expansions in aesthetic phenomena can be a strong driver for a sustainable change in the building industry. In this particular case, the property has shown to appeal to a broad cultural popularity, both with high occupancy and as a treasured visual object in a diverse selection of medias. But the case also shows, that the two-sided concept of spolia being both practical and ideological are not necessarily congruent in regards of the actual environmental impact. This points to even higher demands in architectural design and discussions on ethical use of materials in the early design phases to secure a meaningful use of spolia in new constructions.

5 CONCLUDING REMARKS AND FUTURE PERSPECTIVES

The ambition of this paper has been to investigate and broaden the concept of spolia in connection to buildings from the industrial period to discuss, how a modern concept of spolia

encourage a circular economy by preserving materials, constructions, and questions what kind of architectural strategies can be developed from this?

The case of the Resource Rows and its innovative way of developing new façade motifs based on the recycling of brick fragments from modernist buildings has shown how the concept of spolia is very relevant in this context. The facade becomes an exponent of multiple layers of history. It tells the story of modernist masonry and its limitations while linking the facade to a long-standing Danish masonry tradition. The spolia of the façade becomes a symbol of both the use of masonry in an earlier period and of the need to change today's construction in a more sustainable direction.

The spolia adds value to the façade giving the reused material a symbolic dimension. The tectonic motif of the facade becomes an image of both symbolic historical layers and a solution to today's resource scarcity. But there is a downside to this story, as the chosen solution is inherently unsustainable. The Spolia motif is used to create a sustainable narrative about an unsustainable solution. This points to a potential challenge in consciously working with spolia as architectural motifs. Working with spolia is not sustainable in itself. It needs to be part of a larger, holistic, sustainability strategy to be meaningful in that context.

The future perspectives of working with spolia as part of sustainable architectural strategies calls for deep awareness of both the potentials and challenges of the concept. Today, many projects are working to emphasise the reuse of materials as an important architectural motif. Here, the conscious use of spolia can be an important and productive tool. But perhaps the need for an almost mannered clarification of reuse in architecture is only a temporary phenomenon. Today, the reuse of building materials is something "new" and something you want to emphasise in your design. In the future, it will most likely be an integral part of building and designing and thus the way we work with spolia as a motif may change and become more subtle.

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